

Spatial encoder

spots (p_i, g_i)
graph kernel integral
Gaussian splat
spectral (FNO) blocks

encode
→

Latent field $z(x, y)$

continuous field on
a regular lattice
+ occupancy support

evolve
→

Morphogen operator

$\partial_t z = \nabla \cdot (D_\theta \nabla z) + f_\theta(z)$
Strang splitting;
exact spectral
diffusion solve

decode
→

Decoder

read-out at
any (x, y)
→ $\hat{g}$

trained by masked spatial reconstruction; held-out regions are never seen by the encoder